SYNOPSIS

SMART SALARY PAYROLL SYSTEM

SUBMITTEDIN FULFILLMENTOFTHE
REQUIREMENT FORTHE
AWARD OF THE DEGREE

BACHELORS OF COMPUTER APPLICATIONS

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PROBLEM DEFINITION

In many small and medium-sized organizations, particularly private companies and institutions in India, payroll management is still handled manually using paper records or basic spreadsheets. HR departments spend considerable time calculating salaries, allowances, deductions, and taxes for each employee every month. This traditional process is time-consuming and prone to several inefficiencies.

Manual payroll processing often results in calculation errors, delayed salary payments, and incorrect deductions such as PF, ESI, and TDS. Even small mistakes can create financial discrepancies and dissatisfaction among employees. Maintaining attendance records, overtime details, leave deductions, and bonus calculations manually increases the workload and chances of error.

Additionally, generating monthly payslips, yearly salary summaries, and department-wise reports requires repetitive manual effort. Managing historical records becomes difficult, and retrieving past data for audits or management review can be challenging. Lack of proper data security also puts sensitive employee information at risk.

Therefore, there is a need for an automated Salary Payroll Management System that can ensure accurate salary calculation, reduce manual effort, improve transparency, and maintain secure digital records efficiently.

ACKNOWLEDGEMENT

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INTRODUCTION

Payroll management is one of the most important administrative functions in any organization. It involves calculating employee salaries, managing attendance, applying deductions, and generating payslips.

The proposed Salary Payroll Management System is a web-based application that automates salary processing. The system allows administrators and HR managers to manage employee records, track attendance, calculate gross and net salary, and generate monthly reports.

The system is developed using modern web technologies such as:

- Frontend: React
- Backend: Node.js
- Database: MongoDB, PostgreSQL
- IDE: VS Code
- Browser Support: Latest Brow

The application ensures accuracy, reduces paperwork, improves transparency, and saves time by automating the complete payroll process.

OBJECTIVES

The main objective of this project is to develop a secure and efficient Salary Payroll Management System to automate salary calculations and employee management.

Specific objectives are:

- To create and manage employee records digitally.
- To automate salary calculation including basic pay, allowances, and deductions.
- To calculate Provident Fund (PF), ESI, and TDS automatically.
- To generate monthly payslips in PDF format.
- To maintain salary history for each employee.
- To generate reports such as monthly payroll summary and yearly salary statements.
- To provide secure login for Admin and HR users.
- To reduce manual errors and paperwork.

MODULES

- Administrator Module
- HR Module
- Employee Module

SOFTWARE REQUIREMENT ANALYSIS

Admin Module

Admin can manage the complete payroll system including employees, salary structure, and reports.

- Add / Edit / Delete employees
- Create departments
- Manage salary structure
- Define allowance components (HRA, DA, Bonus)
- Define deduction components (PF, ESI, TDS)
- Generate monthly payroll
- Approve salary processing
- Generate reports (monthly/yearly)
- Export reports in PDF or Excel

HR Module

- Manage employee attendance
- Enter overtime details
- Process salary for selected month
- Generate and download payslips
- View salary history

Employee Module

- Login with username and password
- View monthly payslip
- Download salary slip (PDF)
- View salary history
- View attendance details

HARDWARE AND SOFTWARE SPECIFICATION

HARDWARE REQUIREMENTS:

- **Development System:**
  - **Processor:T2MICRO**
  - **512MB RAM minimum**
  - **10GB Storage**
  - **BUCKET STORAGE: 2GB**
  - **OS: LINUX**
- **Server Requirements:**
  - **512MB RAM minimum**
  - **Cloud Server: AWS**

SOFTWARE REQUIREMENTS:

- **Frontend: React**
- **Backend: Node.js**
- **Database: MongoDB, PostgreSQL**
- **IDE: VS Code**
- **Browser Support: Latest Browsers**

Example: Chrome/Edge/Firefox etc.

Design

Design is the first step in the development phase for defining a system in sufficient detail to permit its physical realization.

The system is divided into three main components:

1. User Interface (Frontend)
2. Application Logic (Backend)
3. Database (Data Storage)

After analyzing the software requirements, system design includes database structure, user interface design, and module flow.

The design ensures reliability, maintainability, scalability, and security of the system.

DATABASE DESIGN

The database stores employee and payroll-related data with minimum redundancy.

Main Tables:

- employees
- departments
- salary_structure
- attendance
- payroll
- payslips
- users

Example Table Structure:

- **Employees:**
  - id
  - employee_id
  - full_name
  - email
  - phone
  - department_id
  - basic_salary
  - created_at
  - updated_at
- **Payroll:**
  - id
  - employee_id
  - month
  - year
  - gross_salary
  - total_deductions
  - net_salary
  - generated_at

THANK YOU